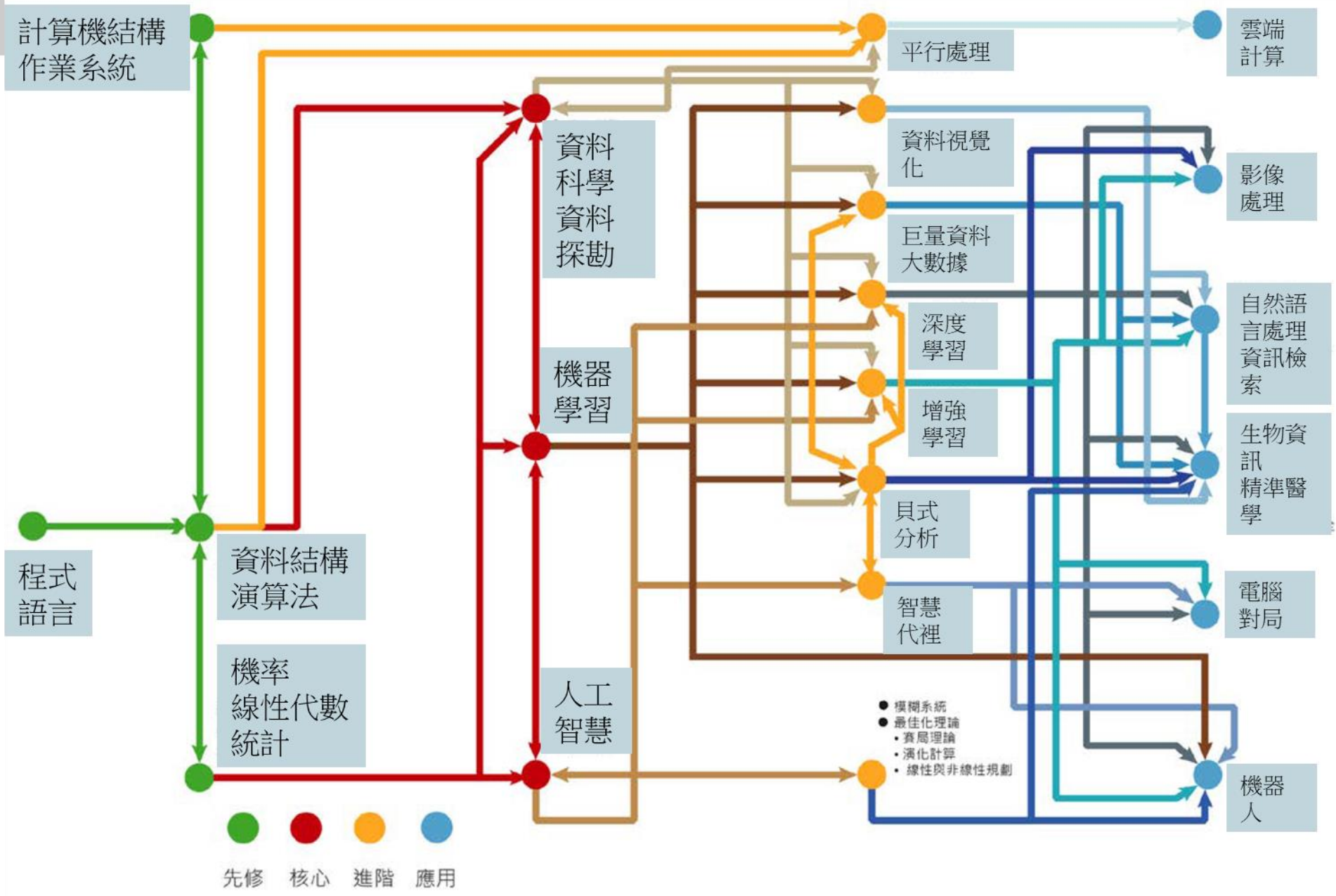




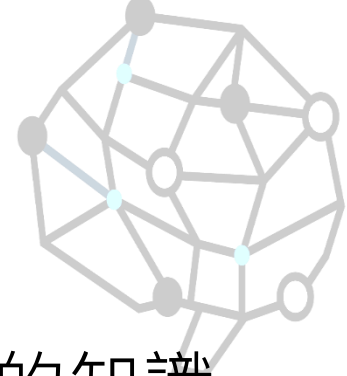
Chap. 0 AI Game

課程：人工智慧導論

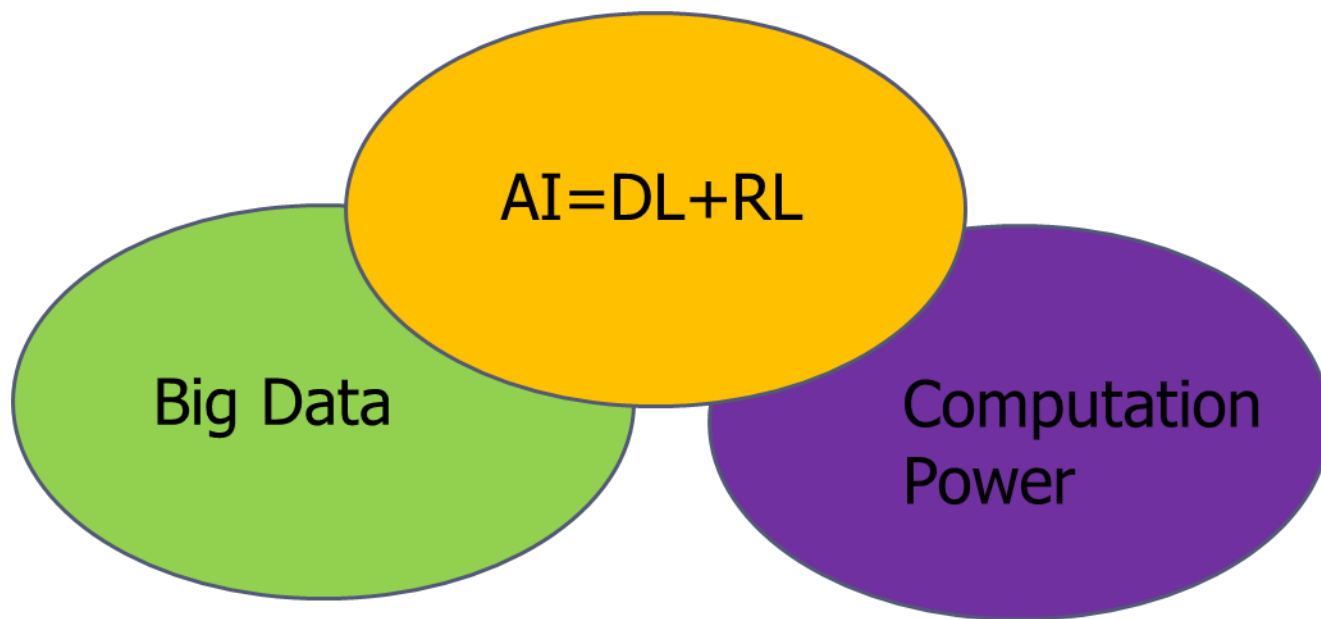
授課教師：周信宏



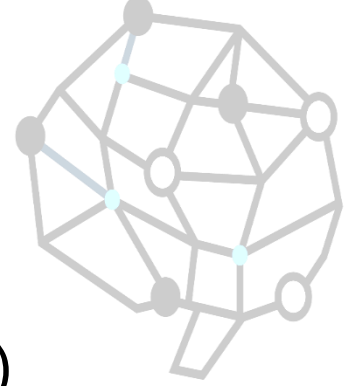
AI=DL+RL(by David Silver)



- Deep Learning : 儲存知識
- Reinforcement Learning : 搜尋得到更好的知識

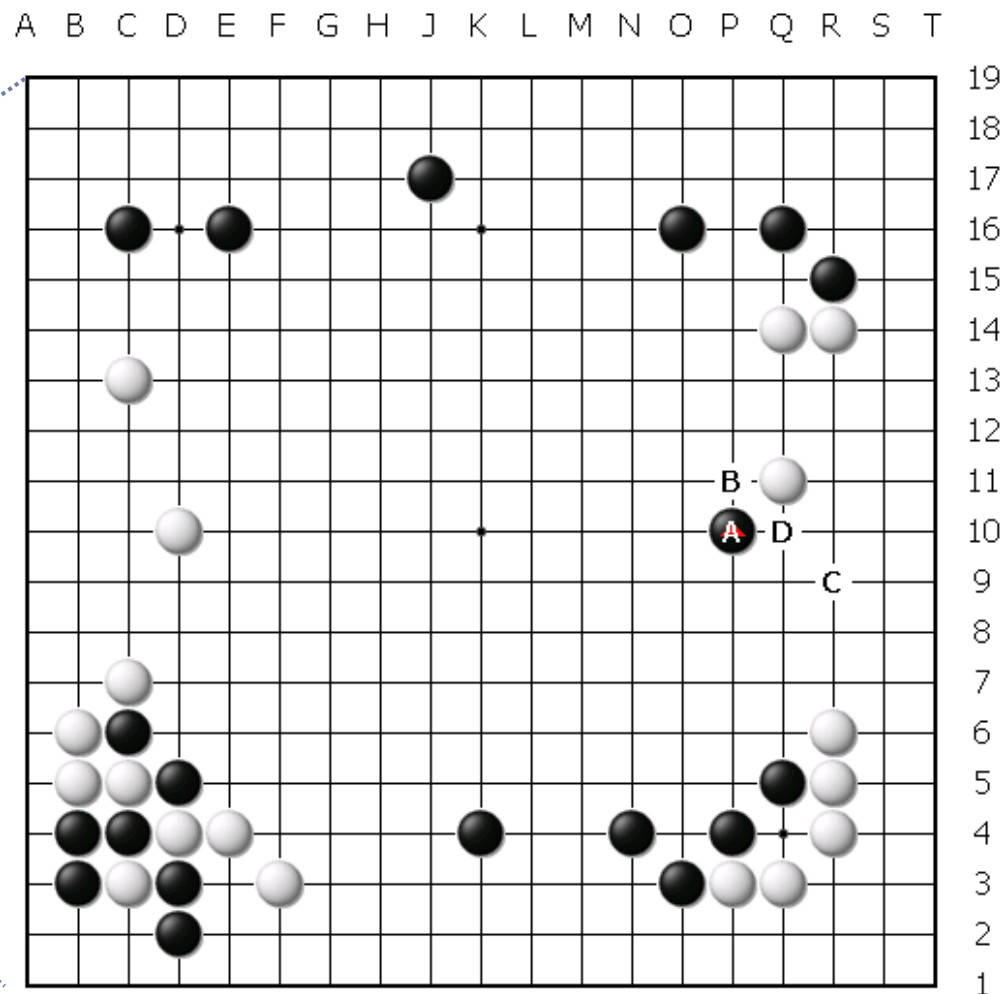
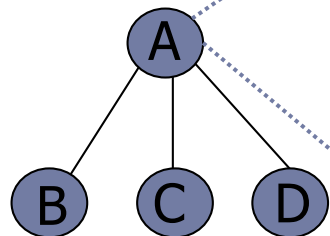


AlphaGo

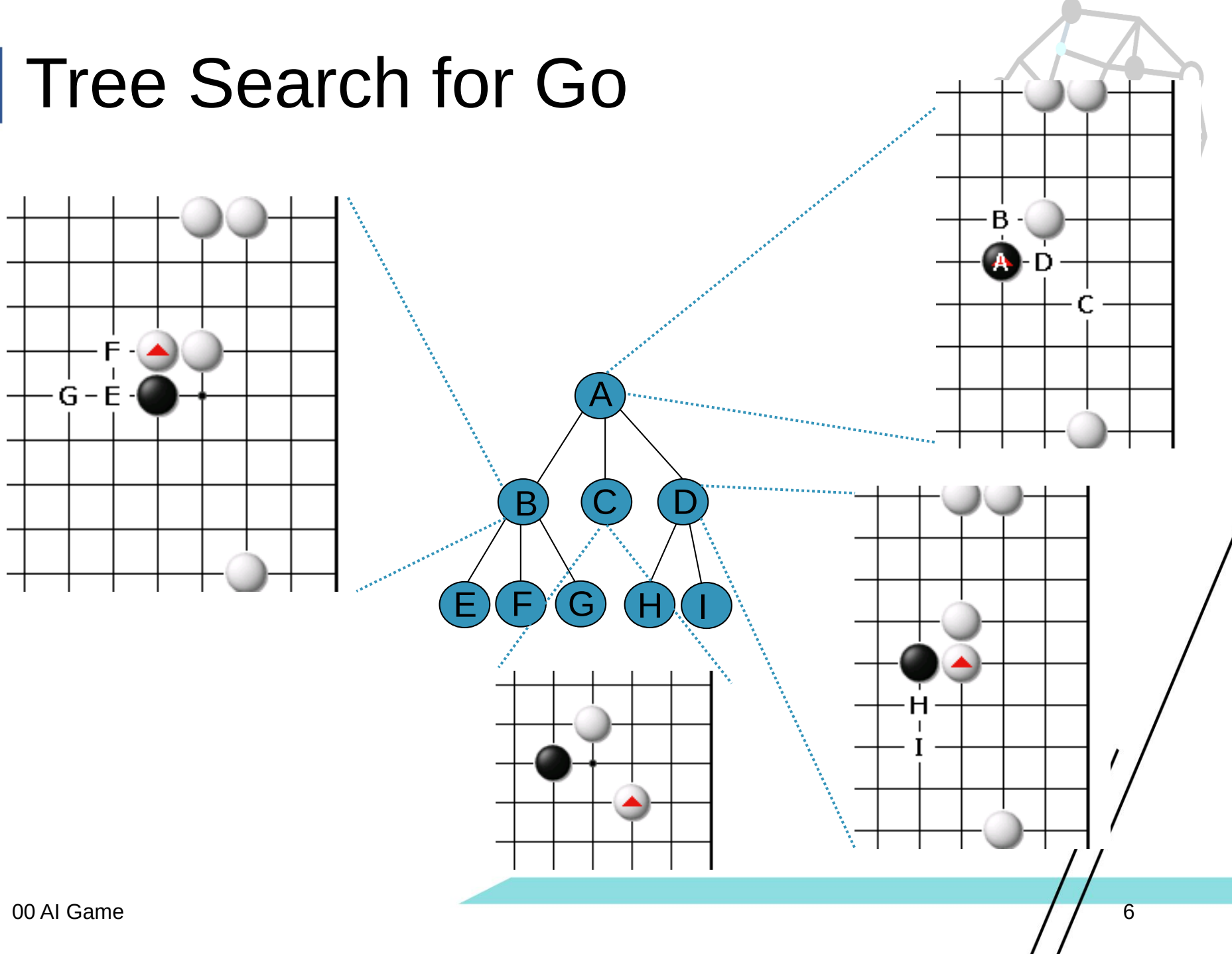


- MCTS + PN + VN + Rollout
- MCTS(Monte Carlo Tree Search, 蒙地卡羅樹搜尋)
 - 思考推理的架構
- Policy Network(策略網路)
 - 記憶學過的棋型，從中找出目前策略
- Value Network (評估網路)& Rollout policy(隨機策略)
 - 形勢判斷

樹狀搜尋範例



Tree Search for Go



甚麼是PN, VN的N?

- Deep learning Network(深度學習類神經網路)

- Multi-class classifier(分類器)

- Input image pixel

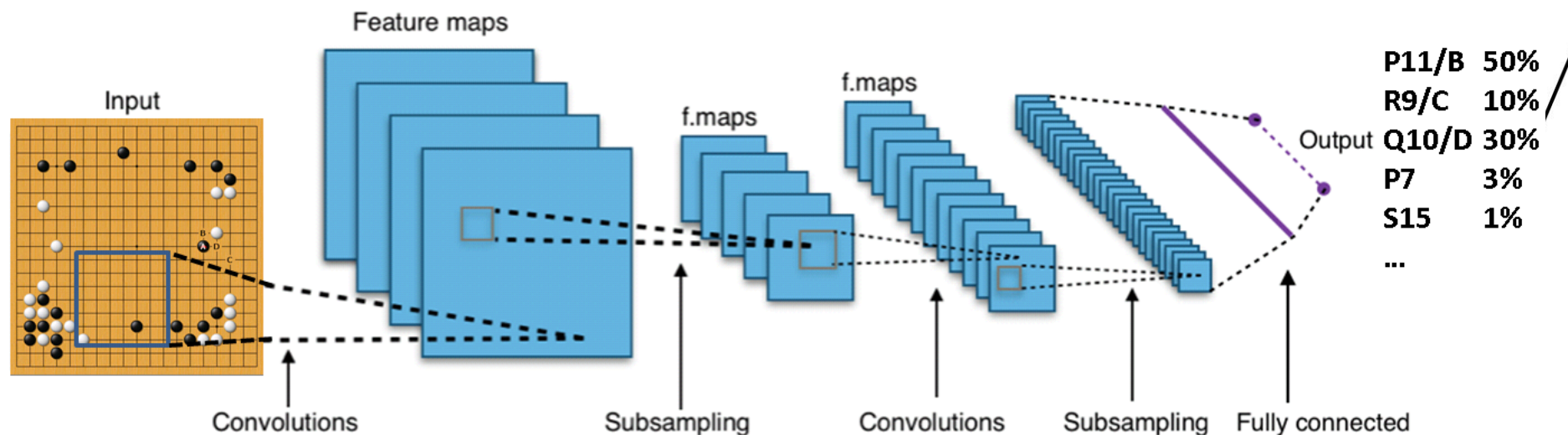
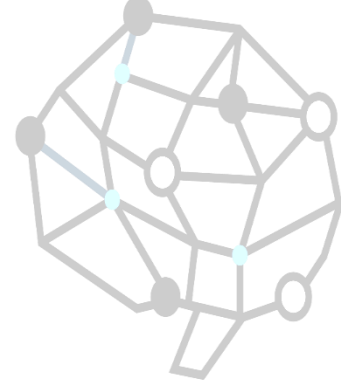
- What are classes' possibilities

- 我想知道圖中有什麼？

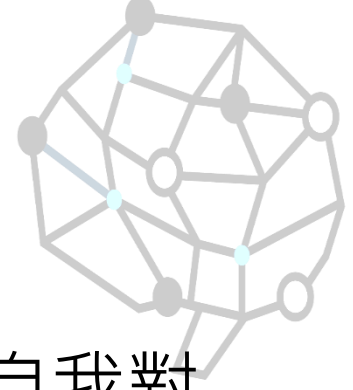


PN(策略網路)

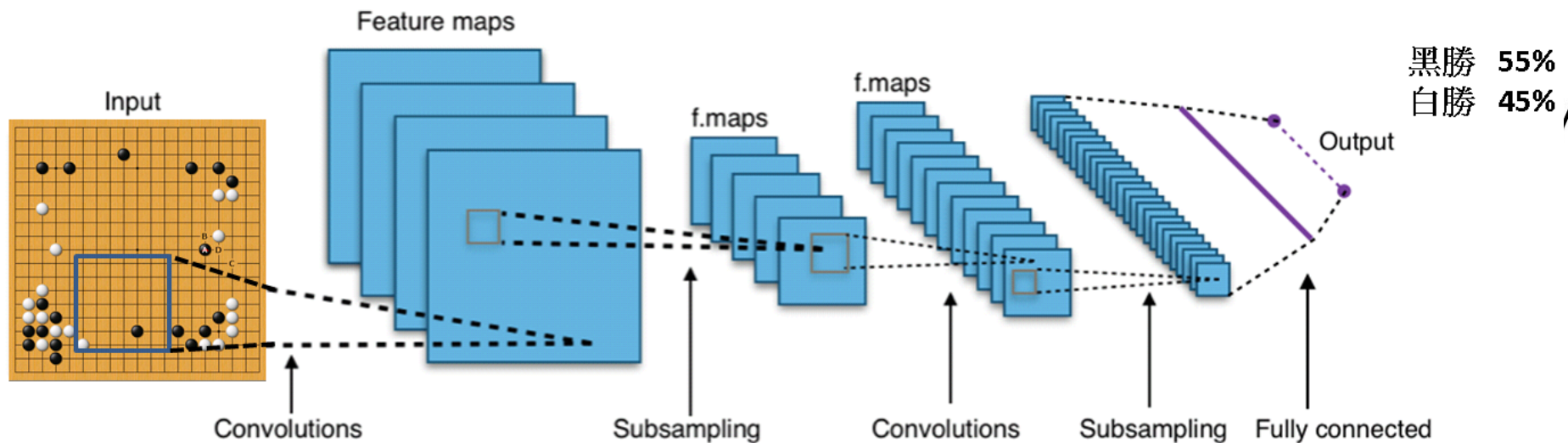
- 我想知道那個著手是好棋
- Training data: KGS 3千萬個盤面



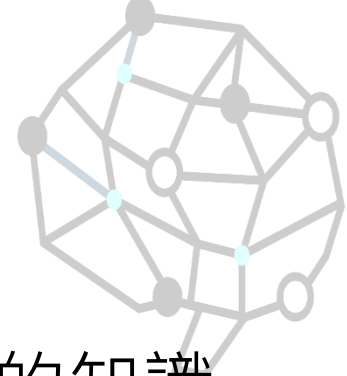
Value Network (評估網路)



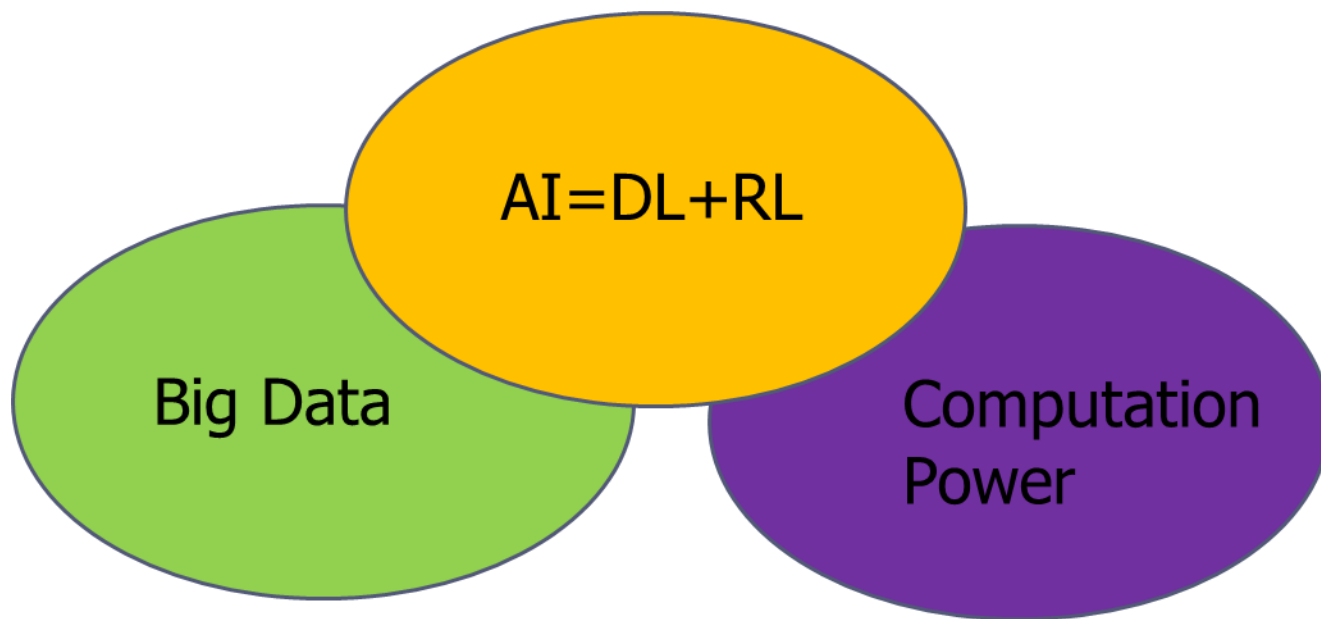
- 我想做形勢判斷
- Training data: 3千萬個盤面，從3千萬盤自我對局取得



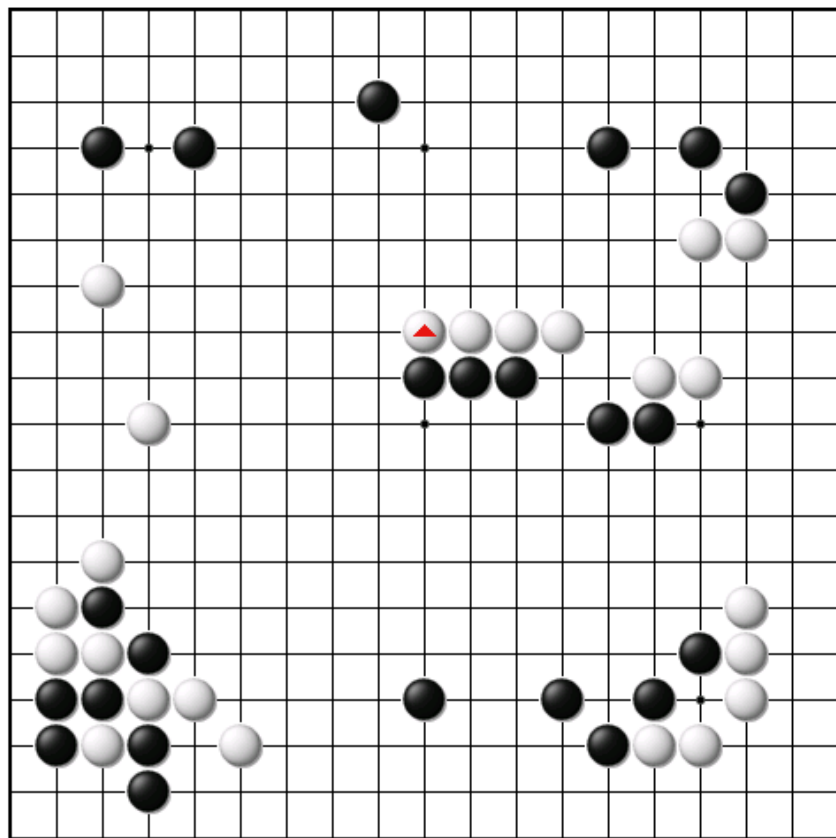
AI=DL+RL(by David Silver)



- Deep Learning : 儲存知識
- Reinforcement Learning : 搜尋得到更好的知識



AlphaGo運作圖解



卡羅樹

最手 → 判盤

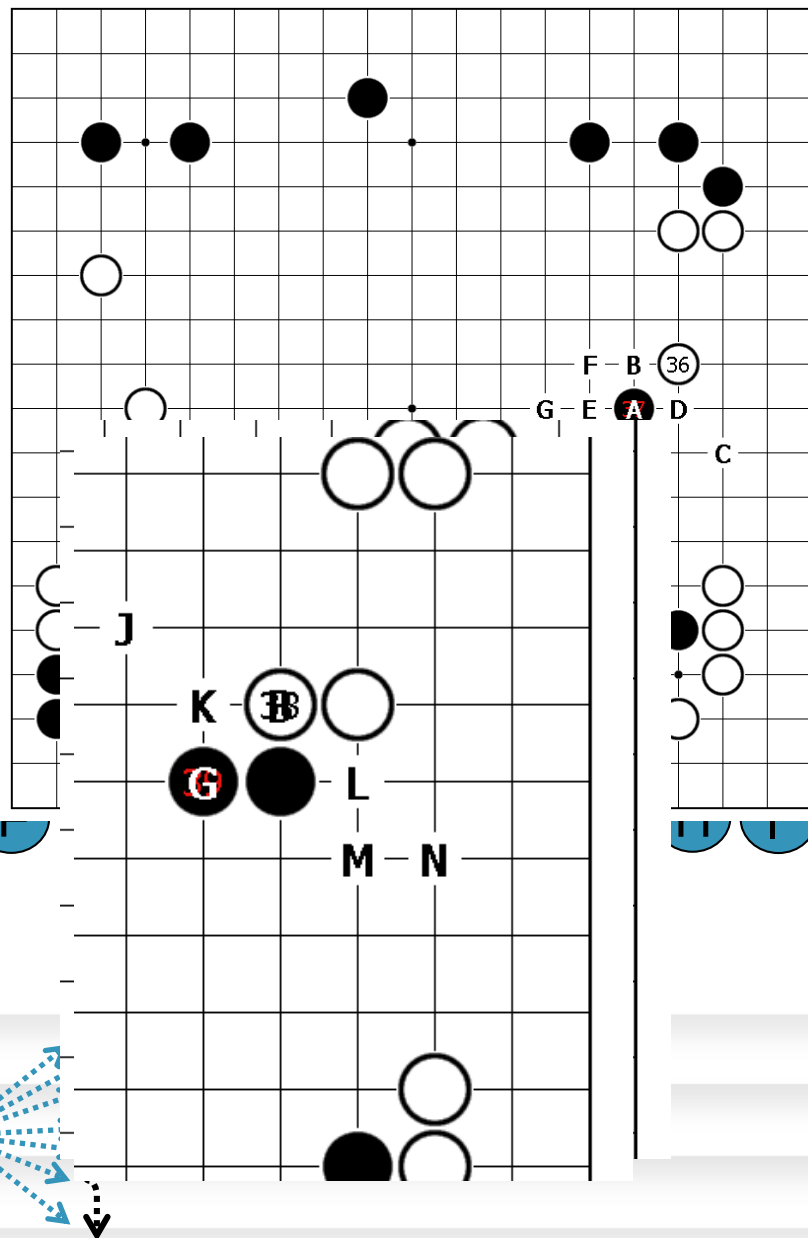
三, ABF,
ADH,

化圖時
擇其中
一手

I E

0.03

MC Rollout



A blue-tinted background image showing a hand moving a chess piece on a chessboard. The text "THE END." is centered over the image.

THE END.
